

Tutorial Problems – MASM File I/O

Problem 1: Line-Number Copy (Basic File I/O)

Write a MASM program using model small and DOS INT 21h services that reads a file named INPUT.TXT and writes its contents to OUTPUT.TXT. Each line in the output file must be prefixed with its line number followed by a colon and a space.

Example:

```
INPUT.TXT:
hello world
asm lab
```

```
OUTPUT.TXT:
1: hello world
2: asm lab
```

Problem 2: Per-Student Grade Files

A file named RECS.TXT contains repeated records in the format:

```
Subject: <subject>
Student: <student_id>
Marks: <marks>
```

For each record, create (or append to) a file named student.TXT and write a line of the form: <subject>:<marks> Each record must be appended as a new line.

Example input:

```
Subject: Math
Student: s1
Marks: 85
```

```
Subject: Physics
Student: s2
Marks: 78
```

```
Subject: Chemistry
Student: s1
Marks: 92
```

Expected files:

s1.TXT:
Math:85
Chemistry:92

s2.TXT:
Physics:78

Example 2

Input file (RECS.TXT):

Subject: English
Student: s3
Marks: 88

Subject: Math
Student: s1
Marks: 91

Subject: Biology
Student: s3
Marks: 73

Subject: Physics
Student: s2
Marks: 84

Subject: History
Student: s1
Marks: 76

Expected output files:

s1.TXT:
Math:91
History:76

s2.TXT:
Physics:84

s3.TXT:
English:88
Biology:73